your application or as a full-fledged framework for building complex single-page applications (SPAs). Vue.js is known for its simplicity, flexibility, and ease of integration with other libraries or existing projects.

In this hands-on lab, we will develop a full stack web application using **Laravel 12** as the backend and **Vue.js** as the frontend. The backend will expose a RESTful API to manage product data, and the frontend will be built using Vue.js with the official Laravel 12 starter kit. The database used is SQLite for simplicity and local development.

15.7.2 Objectives

By the end of this lab, learners will be able to:

- Scaffold a Laravel 12 + Vue.js project using the Laravel CLI
- Develop RESTful API endpoints for Product CRUD operations
- Build a Vue.js interface to consume the Laravel API
- Perform Create, Read, Update, and Delete operations from the Vue UI
- Connect frontend and backend using `axios`

15.7.3 Prerequisites

Ensure the following are installed on your system:

- PHP ≥ 8.2
- Composer
- Laravel Installer
- Node.js and npm
- SQLite installed
- A code editor (e.g., Visual Studio Code)

If you use **Visual Studio Code**, you can install the **Vue - Official** extension for better syntax highlighting and code completion.

15.7.4 Steps

Here are the steps to create a full stack product CRUD application using Laravel 12 and Vue.js.

15.7.4.1 Step 1: Create Laravel 12 Project with Vue.js Starter Kit